

Khac-Hoang Ngo

Assistant Professor

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[Website](#) | [Google Scholar](#) | [ORCID](#) | [GitHub](#) | [LinkedIn](#)

Education and Qualifications

- **Docent in Electrical Engineering** Jun. 2026
Linköping University, Sweden
- **PhD in Wireless Communications** 2020
CentraleSupélec, Paris-Saclay University, France
Thesis: Noncoherent Wireless Communications: Fundamental Limits and System Design ([Link](#), [Video](#))
Supervisors: [Prof. Sheng Yang](#), [Dr. Maxime Guillaud](#)
- **MSc in Wireless Communications** 2016
CentraleSupélec, Paris-Saclay University, France
Thesis: Performance Analysis of Coded Caching ([Link](#))
Supervisors: [Prof. Mari Kobayashi](#), [Prof. Sheng Yang](#)
- **BEng in Electronics and Telecommunications** 2014
Univ. of Engineering and Technology (UET), Vietnam National University Hanoi (VNU)
Thesis: Software-Defined-Radio Implementation of OFDM-based Physical-Layer Network Coding ([Link](#))
Supervisors: [Assoc. Prof. Nguyen Linh Trung](#), [Assoc. Prof. Nguyen Quoc Tuan](#)

Work Experience

Academic Appointments

- **Assistant Professor**, Linköping University, Sweden Sep. 2024–present
- **Postdoctoral Researcher**, Chalmers University of Technology, Sweden Sep. 2020–Aug. 2024
- **Doctoral Researcher**, CentraleSupélec, Paris-Saclay University, France Jul. 2017–Jun. 2020
- **Research Assistant**, CentraleSupélec, Paris-Saclay University, France Feb. 2016–Oct. 2016
- **Research Assistant**, Vietnam National University Hanoi Jul. 2014–Aug. 2015

Industry and Other Experience

- **Research Engineer**, Huawei Paris Research Center, France Nov. 2016–Jun. 2020
- **Intern**, Vietnam Posts and Telecommunications Group Dec. 2013–Mar. 2014
- **Intern**, National University of Singapore Jul. 2012–Aug. 2012

Selected Honors and Awards

- **Golden Globe Award in Science and Technology** for under-35 Vietnamese researchers 2024
- **Best Paper Award**, IEEE Statistical Signal Processing Workshop (SSP) 2023
- **Featured in the spotlight** of the 7th and 9th Heidelberg Laureate Forum, Germany 2019, 2022
- **Marie Skłodowska-Curie Actions Individual Fellowship** 2021

- **Best Paper Award**, Int. Conf. on Advanced Technologies in Communications (ATC) 2021
- **Signal, Image & Vision PhD Thesis Prize** by EEA, GRETSI and GdR-ISIS, France 2021
- **“Impact Science” Second Prize** for PhD thesis, CentraleSupélec Foundation, France 2021
- **Graduated with first-class honors** at both bachelor’s and master’s levels 2014, 2016
- **Honda Young Engineers and Scientists Award**, Honda Foundation, Japan 2013

—— Research Profile

- **Areas of Interest:** Wireless Communications, Information Theory, Machine Learning
Main topics: massive random access, semantic point-cloud transmission, information freshness, AI privacy and security, MIMO, noncoherent communications, coded caching, network coding
- [Google Scholar profile](#), as of Oct. 2026: #citations: **711**, h-index: **13**, i10-index: **20**
- **Publications:** **15** journal papers (10 lead-authored), **37** conference papers (25 lead-authored), **1** book, **1** patent
- [List of publications](#) at the end of the CV

—— Research Funding

Proposal-development role: **lead; * contributor.

- ****Memory Poisoning in LLM Agents: Foundations, Attacks, and Defenses** 2026–2030
Funder: Swedish Research Council (VR)
Amount: SEK 6 million, Role: PI
- ****Point-Cloud Transmission for Remote Registration** 2025–2029
Funder: Wallenberg AI, Autonomous Systems and Software Program (WASP), Sweden
Amount: SEK 8 million, Role: PI
- ***Theory for the Privacy-Security Trade-off in *Practical* Federated Learning** 2025–2029
Funder: Swedish Research Council (VR)
Amount: SEK 5 million, Role: co-PI
- ***Theory for the Privacy-Security Trade-off in Federated Learning** 2023–2027
Funder: Wallenberg AI, Autonomous Systems and Software Program (WASP), Sweden
Amount: SEK 4 million, Role: co-PI
- ****[LANTERN: Low-latency and private edge computing in random-access networks](#)** 2021–2023
Funder: Marie Skłodowska-Curie Individual Fellowship
Amount: EUR 200,000, Role: Fellow
- ***Agricultural Internet of Things Based on Edge Computing** 2022–2024
Funder: ICT Virtual Organization of ASEAN Institutes and Japan’s NICT
Amount: USD 80,000, Role: co-PI
- ***[Connecting the Unconnected: A Tool for Digital Inclusion](#)** 2021–2022
Funder: AlumNode Funding, Klaus Tschira Foundation, Germany
Amount: EUR 5,000, Role: co-PI

Invited Talks by Topic

Massive Random Access

- VinUniversity, Vietnam Jul. 2026
- Universitat Politècnica de Catalunya, Spain Jun. 2026
- University of Cyprus Oct. 2025
- Swedish Communication Technologies Workshop (1.5-hour tutorial) Oct. 2025
- Aalborg University Apr. 2024
- Linköping University Oct. 2023
- Vietnam National University Hanoi May 2023, May 2021
- INRIA Lyon Apr. 2023
- CentraleSupélec Jan. 2023
- Zugspitze Workshop on Communications Jan. 2023
- Equipe Traitement de l'Information et Systèmes (ETIS), France Feb. 2022
- H2020 INCOMING Summer School (1.5-hour tutorial) Jun. 2022

Age of Information

- Association of Vietnamese Science and Technology Experts in Sweden Nov. 2024
- University of Amsterdam Apr. 2024
- Hanoi University of Science and Technology Dec. 2023
- German Aerospace Center (DLR) Sep. 2022

Noncoherent Wireless Communications

- 8th Francophone Symposium on Signal and Image Processing (GRETSI), France Sep. 2022
- CentraleSupélec Dec. 2021, Jun. 2018
- Club EEA, France Jun. 2021
- Vietnam National University Hanoi Aug. 2019, Nov. 2017
- Technical University of Munich Sep. 2017

Coded Caching

- Vietnam National University Hanoi Aug. 2016

Higher-Education Pedagogical Training

- **Higher Education Pedagogy for Doctoral Supervisors and Course Coordinators** 2025
Linköping University, Sweden
- **Diploma in Teaching and Learning in Higher Education** 2024
Chalmers University of Technology, Sweden

Supervision

Doctoral Students: Principal Supervisor

- **Lorenzo Sucameli**, Linköping University, Sweden Starting Nov. 2026
Topic: Memory poisoning in LLM agents
- **Garima Bazard**, Linköping University, Sweden Oct. 2025–present
Topic: Point cloud transmission for remote registration
- **Shipeng Liu**, Linköping University, Sweden Aug. 2025–present
Topic: Point cloud transmission for remote registration

Doctoral Students: Co-Supervisor

- **Siying Chen**, Linköping University, Sweden Jun. 2026–present
Topic: Distributed inference
- **Amandus Reimer**, Chalmers University of Technology, Sweden Mar. 2025–present
Topic: AI security
- **Marcus Lassila**, Chalmers University of Technology, Sweden Sep. 2023–present
Topic: Membership inference attack

Master's Theses and Projects: Co-Supervisor

- **Xi Zhang**, Chalmers University of Technology, Sweden Sep. 2023–Aug. 2024
Topic: Learning joint detection and decoding in short-packet communications
Now a doctoral student at Tampere University, Finland
- **Khodor Safa** and **Shanglin Yang**, CentraleSupélec, France Mar. 2021
Topic: MIMO detection under the generalized Gaussian model
- **Wassim Khelil, Mohamed Idriss Khaledi, Anas Ouallou**, CentraleSupélec, France Mar. 2020
Topic: Embracing non-linearities in future wireless systems via non-convex optimization

Bachelor's Theses: Principal Supervisor

- **Noa Paguera** (Erasmus exchange student), Linköping University, Sweden Aug. 2025–Jan. 2026
Topic: Knowledge distillation for light-weight smart wireless devices

Teaching

Course Instructor and Examiner, Linköping University

- **TSKS18 Generative AI for Data Compression and Transmission** Fall 2026
Master's-level course, 6 credits. I initiated and developed the course.
- **TSKS36 Digital and Wireless Communications** Spring 2026
Master's-level course, 6 credits
- **TSIN01 Information Networks** Fall 2025
Master's-level course, 6 credits

Teaching Assistant, Chalmers University of Technology

- **SSY135 Wireless Communications** Spring 2023, 2024
Master's-level course, 7.5 credits, ≈ 20 students
- **SSY210 Information Theory** Spring 2021, 2023
Master's- and doctoral-level course, 7.5 credits, ≈ 20 students
- **EEN110 High-Dimensional Statistics** Fall 2021, 2023
Master's- and doctoral-level course, 7.5 credits, ≈ 12 students

Guest Lectures

- **Modern random access protocols: coded slotted ALOHA** Fall 2024, 2026
In *TSIN01 Information Networks*, Linköping University

Professional Service and Activity

Professional Membership and Leadership Positions

- **Member:** IEEE (since 2017); ACM (2022–2023); AVISE, Association of Vietnamese Science and Technology Experts in Sweden (since 2024)
- **Chair, IEEE Information Theory Society (ITSoc) Student and Outreach Subcommittee** 2026–2027
Responsible for organizing social events and student competitions at ISIT, ITW, and other ITSoc conferences
- **Founding member and administrator** of telecom-vn, a Facebook group for Vietnamese researchers in telecommunications. Organize [online seminars](#) and maintain research discussions.

Editorial Roles

- **Associate Editor**, IEEE Wireless Communications Letters Mar. 2025–present
- **Lead Editor**, [IEEE Communications Society Young Professionals Blog](#) Feb. 2025–present
- **Copyeditor**, ICT Research Journal, Vietnam Ministry of Information and Communications 2021

Evaluation and Reviewing

- **Member of degree examination committees**
 - **Aniol Martí Espelt**, Universitat Politècnica de Catalunya, Spain Jun. 2026
PhD thesis, *Physically consistent wireless communications with statistical channel state information*
 - **Diego Cuevas Fernández**, University of Cantabria, Spain Nov. 2024
PhD thesis, *Advanced Grassmannian Constellation Designs for Noncoherent MIMO Communications*
 - **Andrea Ramos**, Universitat Politècnica de València, Spain (external reviewer) 2024
PhD thesis, *6G Integrated Sensing and Communication System for the Factory of the Future*
 - **Lorentz Henriksson**, Linköping University 2025
Master's thesis, *Orientation estimation in 6G radio networks using artificial neural networks*
- **Reviewer for research grants**
 - National Foundation for Science and Technology Development (NAFOSTED), Vietnam 2023
- **Regular reviewer for 17 international journals**, including IEEE TIT, TWC, TCOM, and TSP
- **Regular reviewer for major international conferences**, including IEEE ISIT, ICC, GLOBECOM, SPAWC, and WCNC

Conference Leadership

- **Conference organization: major roles**
 - **Publication Chair**, IEEE Inf. Theory Workshop (ITW) 2028
 - **Lead organizer**, ELLIIT Focus Period 2027
Theme: Cooperative Predictive Planning Toward Vision Zero
 - **Communication track chair**, Int. Conf. Adv. Techno. Commun. (ATC) 2022
 - **Scientific committee member**, 1st Junior Conf. in Wireless & Optical Commun. 2019
Paris-Saclay University, France
 - **Executive committee member**, 1st Honda Forum for Young Engineers and Scientists 2015
Tokyo, Japan

- **Conference organization: session/workshop organizer**
 - Special session: *Predictive intelligence for next-generation wireless networks* 2026
[Asilomar Conference on Signals, Systems, and Computers](#)
 - Special session: *Massive random access* 2026
IEEE Int. Workshop on Signal Processing and AI in Wireless Communications (SPAWC)
 - Workshop: *Resource allocation and cooperation in wireless networks (RAWNET)* 2025
Int. Symp. on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt)
 - Hot-topic panel discussion: *Connecting the Unconnected* 2022
IEEE Int. Symp. World of Wireless, Mobile Multi. Netw. (WoWMoM), Belfast, UK
 - Special session: *Noncoherent communications* 2021
25th Int. ITG Workshop Smart Antennas (WSA), France
 - Special session: *Edge computing for 5G and beyond-5G networks* 2021
Int. Conf. Advanced Techno. Commun. (ATC), Vietnam
- **Conference organization: session chair**
 - *Massive multiple access: rate regions and interference*, IEEE ISIT 2023
 - *Age of information*, IEEE GLOBECOM 2023
- **Technical program committee member**
 - IEEE Int. Symp. on Information Theory (ISIT) 2026
 - IEEE Global Commun. Conf. (GLOBECOM) 2026
 - IEEE Int. Conf. Comm. (ICC) 2026
 - IEEE Wireless Commun. Netw. Conf. (WCNC) 2024, 2025
 - IEEE Int. Symp. Personal, Indoor and Mobile Radio Commun. (PIMRC) 2025
 - Joint EuCNC & 6G Summit 2023, 2025
 - Int. Conf. Control, Auto. Inf. Sciences (ICCAIS) 2023
 - APSIPA Annual Summit and Conference 2023
 - Int. Symp. Inf. Commun. Techno. (SoICT) 2022, 2023
 - Int. ITG Workshop Smart Anten. (WSA) 2023

Outreach

- **Science communicator** for the Vietnamese community
Organized online workshops, gave public science lecture, e.g., “How to keep information fresh?”, “Wireless Communications: Basics and Applications”
- **Participant in scientific forums**
 - 7th, 9th, and 13th [Heidelberg Laureate Forum](#), Germany 2019, 2022, 2026
 - 2nd and 3rd Global Young Vietnamese Scholars Forum, Vietnam Nov. 2019, Nov. 2020
 - [Honda Young Engineers and Scientists \(Y-E-S\) Forum](#), Tokyo, Japan Nov. 2015
 - Young Engineers and Scientists Tokyo Meeting, Tokyo, Japan Nov. 2014
- **Media coverage**
 - [Spotlight interview at the 9th Heidelberg Laureate Forum](#) about my research and challenges in telecommunications
 - [Interview with the French National Doctoral Network](#) about my PhD journey
 - “10 out of 200: Serving the People” profile for the 7th Heidelberg Laureate Forum, featuring my research in wireless communications
 - Various Vietnamese newspapers

Languages

Vietnamese (native), English (fluent), French (elementary), Swedish (elementary)

Publications

Preprints/Submitted

- [Pre1] M. Lassila, J. Östman, **K.-H. Ngo**, and A. Graell i Amat, “Hierarchical Bayesian membership inference: A general framework with applications to diffusion models,” in *Submitted to Proc. Int. Conf. Learn. Represent. (ICLR)*, 2027.
- [Pre2] S. Liu, G. Bazard, **K.-H. Ngo**, and P.-E. Forssén, “Point cloud transmission with patch-based rate adaptation,” in *Submitted to Proc. IEEE Int. Conf. Commun. (ICC)*, Washington, DC, USA, 2027.
- [Pre3] J. Östman, M. Lassila, **K.-H. Ngo**, and A. Graell i Amat, “Predictive geometry in language models,” in *Submitted to Proc. Int. Conf. Learn. Represent. (ICLR)*, 2027.
- [Pre4] A. Lancho, **K.-H. Ngo**, S. Rini, R. Oliver, and R. Vazquez-Alvarez, “Charting the path to Bifröst: The 2026 ESIT Data-to-Information Competition on robust CSI-based localization and machine unlearning,” *Submitted to IEEE BITS Inf. Theory Mag.*, 2026.

Book

- [B1] N. L. Trung, V. N. Q. Bao, N. V. Ha, L. V. Ha, **K.-H. Ngo**, et al., *Two-way relay communications: theory and implementation*. Hanoi, Vietnam: Vietnam National University Publishing House, 2025, Language: Vietnamese.

Patent

- [P1] **K.-H. Ngo**, A. Decurninge, M. Guillaud, and S. Yang, “Transmitter and receiver communication apparatus for non-coherent communication,” U.S. Patent 17/243,679, Aug. 19, 2021.

Journal Papers

- [J1] J.-F. Chamberland, K. Narayanan, **K.-H. Ngo**, G. Durisi, V. T. Kunde, and C. S. K. Valmeekam, “Prediction-based compression using large language models,” *IEEE BITS Inf. Theory Mag.*, Aug. 2026, Early Access.
- [J2] **K.-H. Ngo**, D. Cuevas, R. de Miguel Gil, V. Monzon Baeza, A. Garcia Armada, and I. Santamaria, “Noncoherent MIMO communications: Theoretical foundation, design approaches, and future challenges,” *IEEE Open J. Veh. Technol.*, pp. 381–401, 2026. [Preprint](#).
- [J3] **K.-H. Ngo** and E. G. Larsson, “Distributed MIMO with over-the-air phase calibration integrated into the TDD flow,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 17 463–17 478, 2026. [Preprint](#).
- [J4] K. Okumus, **K.-H. Ngo**, G. Durisi, and E. G. Ström, “Type-based unsourced multiple access over fading channels in distributed MIMO with application to multi-target localization,” *IEEE Trans. Wireless Commun.*, pp. 22 339–22 353, 2026. [Preprint](#), [Code](#).
- [J5] **K.-H. Ngo**, G. Durisi, A. Munari, F. Lázaro, and A. Graell i Amat, “Timely status updates in slotted ALOHA networks with energy harvesting,” *IEEE Trans. Commun.*, vol. 73, no. 9, pp. 7288–7303, Sep. 2025. [Preprint](#), [Code](#).
- [J6] **K.-H. Ngo**, G. Durisi, A. Graell i Amat, P. Popovski, A. E. Kalor, and B. Soret, “Unsourced multiple access with common alarm messages: Network slicing for massive and critical IoT,” *IEEE Trans. Commun.*, vol. 72, no. 2, pp. 907–923, Feb. 2024. [Preprint](#).
- [J7] **K.-H. Ngo**, A. Lancho, G. Durisi, and A. Graell i Amat, “Unsourced multiple access with random user activity,” *IEEE Trans. Inf. Theory*, vol. 69, no. 7, pp. 4537–4558, Feb. 2023. [Preprint](#), [Code](#).
- [J8] G. Gur, A. Kalla, C. de Alwis, Q.-V. Pham, **K.-H. Ngo**, M. Liyanage, and P. Porambage, “Integration of ICN and MEC in 5G and beyond networks: Mutual benefits, use cases, challenges, standardization, and future research,” *IEEE Open J. Commun. Soc.*, vol. 3, pp. 1382–1412, Aug. 2022.

- [J9] **K.-H. Ngo**, S. Yang, M. Guillaud, and A. Decurninge, “Joint constellation design for noncoherent MIMO multiple-access channels,” *IEEE Trans. Inf. Theory*, vol. 68, no. 11, pp. 7281–7305, Jul. 2022. [Preprint](#).
- [J10] A. U. Rahman, F. Fourati, **K.-H. Ngo**, A. Jindal, and M.-S. Alouini, “Network graph generation through adaptive clustering and infection dynamics: A step towards global connectivity,” *IEEE Commun. Lett.*, vol. 26, no. 4, pp. 783–787, Jan. 2022. [Preprint](#).
- [J11] F. Zhang, **K.-H. Ngo**, S. Yang, and A. Nosratinia, “Transmit correlation diversity: Generalization, new techniques, and improved bounds,” *IEEE Trans. Inf. Theory*, vol. 68, no. 6, pp. 3841–3869, Jan. 2022, (Zhang and Ngo contributed equally to the technical content). [Preprint](#).
- [J12] **K.-H. Ngo**, A. Decurninge, M. Guillaud, and S. Yang, “Cube-Split: A structured Grassmannian constellation for non-coherent SIMO communications,” *IEEE Trans. Wireless Commun.*, vol. 19, no. 3, pp. 1948–1964, Mar. 2020. [Link](#), [Code](#).
- [J13] **K.-H. Ngo**, M. Guillaud, A. Decurninge, S. Yang, and P. Schniter, “Multi-user detection based on expectation propagation for the non-coherent SIMO multiple access channel,” *IEEE Trans. Wireless Commun.*, vol. 19, no. 9, pp. 6145–6161, Sep. 2020. [Preprint](#).
- [J14] T.-T.-Q. Tran, L. V. Nguyen, **K.-H. Ngo**, L.-T. Nguyen, Q.-T. Nguyen, N.-Q.-B. Vo, X.-N. Tran, E. Bastug, S. Azarian, M. Debbah, and P. Duhamel, “Network coding with multimedia transmission and cognitive networking: An implementation based on software-defined radio,” *REV Journal on Electronics and Communications*, vol. 10, no. 3-4, pp. 72–84, 2020, **Invited Article**. [Preprint](#).
- [J15] **K.-H. Ngo**, S. Yang, and M. Kobayashi, “Scalable content delivery with coded caching in multi-antenna fading channels,” *IEEE Trans. Wireless Commun.*, vol. 17, no. 1, pp. 548–562, Jan. 2018. [Link](#).

Conference Papers

- [C1] D. P. Krishnan, K. Okumus, **K.-H. Ngo**, and G. Durisi, “Minimum energy per bit of unsourced multiple access with location-based codebook partitioning,” in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Guangzhou, China, 2026. [Preprint](#).
- [C2] S. Liu, Z. Xiong, **K.-H. Ngo**, and P.-E. Forssén, “MATTER: Multiscale attention for registration error regression,” in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, Barcelona, Spain, 2026. [Preprint](#).
- [C3] K. Okumus, **K.-H. Ngo**, U. K. Ganesan, G. Durisi, E. G. Ström, and S. R. Pandey, “Type-based unsourced federated learning with client self-selection,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, Glasgow, Scotland, UK, 2026. [Preprint](#), [Code](#).
- [C4] N.-S. Duong, **K.-H. Ngo**, T.-M. Dinh-Thi, and V.-L. Nguyen, “Sparse orthogonal matching pursuit-based parameter estimation for integrated sensing and communications,” in *IEEE INFOCOM Workshop*, London, UK, 2025. [Preprint](#).
- [C5] **K.-H. Ngo**, G. Durisi, and P. Popovski, “Information age and correctness for energy harvesting devices with random access,” in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Taipei, Taiwan, 2025, pp. 5411–5416. [Preprint](#).
- [C6] **K.-H. Ngo** and E. G. Larsson, “Breaking the TDD flow for over-the-air phase synchronization in distributed antenna systems,” in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Taipei, Taiwan, 2025, pp. 1645–1650. [Preprint](#).
- [C7] **K.-H. Ngo**, D. N. Nguyen, and T.-M. D. Thi, “Protocol design for irregular repetition slotted ALOHA with energy harvesting to maintain information freshness,” in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Milan, Italy, 2025. [Preprint](#).

- [C8] D. P. Krishnan, K. Okumus, **K.-H. Ngo**, and G. Durisi, “An achievability bound for type-based unsourced multiple access,” in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Ann Arbor, MI, USA, 2025, pp. 1751–1756. [Preprint](#).
- [C9] M. Lassila, J. Östman, **K.-H. Ngo**, and A. Graell i Amat, “Practical Bayes-optimal membership inference attacks,” in *Proc. Conf. Neural Inf. Process. Syst. (NeurIPS)*, San Diego, CA, USA, 2025, pp. 38 522–38 556. [Preprint](#).
- [C10] K. Okumus, **K.-H. Ngo**, G. Durisi, and E. G. Ström, “Type-based unsourced multiple access over fading channels with cell-free massive MIMO,” in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Ann Arbor, MI, USA, 2025, pp. 1518–1523. [Preprint](#), [Code](#).
- [C11] **K.-H. Ngo**, D. P. Krishnan, K. Okumus, G. Durisi, and E. G. Ström, “Type-based unsourced multiple access,” in *Proc. IEEE Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Lucca, Italy, 2024, pp. 911–915. [Preprint](#), [Code](#).
- [C12] **K.-H. Ngo**, J. Östman, G. Durisi, and A. Graell i Amat, “Secure aggregation is NOT private against membership inference attacks,” in *Proc. Eur. Conf. Mach. Learn. Princ. Pract. Knowl. Discov. Databases (ECML PKDD)*, Vilnius, Lithuania, Sep. 2024, pp. 180–198. [Preprint](#), [Code](#).
- [C13] **K.-H. Ngo**, J. Östman, and A. Graell i Amat, “Local mutual-information differential privacy,” in *Proc. IEEE Inf. Theory Workshop (ITW)*, Shenzhen, China, 2024, pp. 31–36. [Preprint](#).
- [C14] N.-S. Duong, Q.-T. Nguyen, **K.-H. Ngo**, and T.-M. Dinh-Thi, “Sparse Bayesian learning with atom refinement for mmWave MIMO channel estimation,” in *Proc. IEEE Stat. Signal Process. Workshop (SSP)*, Hanoi, Vietnam, Jul. 2023, pp. 155–159.
- [C15] **K.-H. Ngo**, G. Durisi, A. Graell i Amat, A. Munari, and F. Lázaro, “Age of information in slotted ALOHA with energy harvesting,” in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Kuala Lumpur, Malaysia, Dec. 2023, pp. 3579–3584. [Preprint](#), [Code](#).
- [C16] **K.-H. Ngo**, A. Graell i Amat, and G. Durisi, “Irregular repetition slotted ALOHA over the binary adder channel,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, Rome, Italy, May 2023, pp. 5346–5351. [Preprint](#).
- [C17] N. T. Nguyen, N. Shlezinger, **K.-H. Ngo**, V.-D. Nguyen, and M. Juntti, “Joint communications and sensing design for multi-carrier MIMO systems,” in *Proc. IEEE Stat. Signal Process. Workshop (SSP)*, **Best Paper Award**, Hanoi, Vietnam, Jul. 2023, pp. 110–114. [Preprint](#).
- [C18] **K.-H. Ngo**, G. Durisi, A. Graell i Amat, P. Popovski, B. Soret, and A. E. Kalør, “Unsourced multiple access for heterogeneous traffic requirements,” in *Proc. Asilomar Conf. Signals Syst. Comput.*, **Invited Paper**, CA, USA, Oct. 2022, pp. 687–691.
- [C19] **K.-H. Ngo**, G. Durisi, and A. Graell i Amat, “Age of information in prioritized random access,” in *Proc. Asilomar Conf. Signals Syst. Comput.*, **Invited Paper**, CA, USA, Oct. 2021, pp. 1502–1506. [Preprint](#), [Video](#), [Code](#).
- [C20] **K.-H. Ngo**, A. Lancho, G. Durisi, and A. Graell i Amat, “Massive uncoordinated access with random user activity,” in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Melbourne, Victoria, Australia, 2021, pp. 3014–3019. [Preprint](#), [Video](#), [Code](#).
- [C21] **K.-H. Ngo**, N. T. Nguyen, T. Q. Dinh, T.-M. Hoang, and M. Juntti, “Low-latency and secure computation offloading assisted by hybrid relay-reflecting intelligent surface,” in *Proc. Int. Conf. Adv. Technol. Commun. (ATC)*, **Best Paper Award**, Hanoi, Vietnam, Oct. 2021, pp. 306–311. [Preprint](#), [Video](#), [Code](#).
- [C22] **K.-H. Ngo** and S. Yang, “A generalized Gaussian model for wireless communications,” in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Melbourne, Victoria, Australia, 2021, pp. 3237–3242. [Preprint](#), [Video](#).

- [C23] **K.-H. Ngo** and S. Yang, “A Riemannian metric for non-coherent constellation design and its application to multiple access channel,” in *Proc. Int. ITG Workshop Smart Antennas (WSA)*, French Riviera, France, Nov. 2021. [Preprint](#).
- [C24] **K.-H. Ngo**, S. Yang, and M. Guillaud, “The optimal DoF for the noncoherent MIMO channel with generic block fading,” in *Proc. IEEE Inf. Theory Workshop (ITW)*, Riva del Garda, Italy, Apr. 2021. [Preprint](#), [Video](#).
- [C25] **K.-H. Ngo**, S. Yang, M. Guillaud, and A. Decurninge, “Noncoherent MIMO multiple-access channels: A joint constellation design,” in *Proc. IEEE Inf. Theory Workshop (ITW)*, Riva del Garda, Italy, Apr. 2021. [Preprint](#), [Video](#).
- [C26] **K.-H. Ngo**, F. Zhang, S. Yang, and A. Nosratinia, “Two-user MIMO broadcast channel with transmit correlation diversity: Achievable rate regions,” in *Proc. IEEE Inf. Theory Workshop (ITW)*, Kanazawa, Japan, Nov. 2021. [Preprint](#), [Video](#).
- [C27] **K.-H. Ngo**, M. Guillaud, A. Decurninge, S. Yang, S. Sarkar, and P. Schniter, “Non-coherent multi-user detection based on expectation propagation,” in *Proc. Asilomar Conf. Signals Syst. Comput.*, CA, USA, Nov. 2019, pp. 2092–2096. [Preprint](#).
- [C28] **K.-H. Ngo**, A. Decurninge, M. Guillaud, and S. Yang, “A multiple access scheme for non-coherent SIMO communications,” in *Proc. Asilomar Conf. Signals Syst. Comput.*, CA, USA, Oct. 2018, pp. 1846–1850. [Preprint](#).
- [C29] **K.-H. Ngo**, S. Yang, and M. Guillaud, “The optimal DoF region for the two-user non-coherent SIMO multiple-access channel,” in *Proc. IEEE Inf. Theory Workshop (ITW)*, Guangzhou, China, Nov. 2018. [Preprint](#).
- [C30] A. Ghorbel, **K.-H. Ngo**, R. Combes, M. Kobayashi, and S. Yang, “Opportunistic content delivery in fading broadcast channels,” in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Singapore, Dec. 2017. [Preprint](#).
- [C31] **K.-H. Ngo**, A. Decurninge, M. Guillaud, and S. Yang, “Design and analysis of a practical codebook for non-coherent communications,” in *Proc. Asilomar Conf. Signals Syst. Comput.*, CA, USA, Oct. 2017, pp. 1237–1241. [Preprint](#).
- [C32] **K.-H. Ngo**, S. Yang, and M. Guillaud, “An achievable DoF region for the two-user non-coherent MIMO broadcast channel with statistical CSI,” in *Proc. IEEE Inf. Theory Workshop (ITW)*, Taiwan, Nov. 2017, pp. 604–608. [Preprint](#).
- [C33] **K.-H. Ngo**, S. Yang, and M. Kobayashi, “Cache-aided content delivery in MIMO channels,” in *Proc. Annu. Allerton Conf. Commun. Control Comput.*, IL, USA, Sep. 2016, pp. 93–100. [Preprint](#).
- [C34] **K.-H. Ngo**, S. Yang, M. Kobayashi, and K. Huang, “On the complementary roles of massive MIMO and coded caching for content delivery,” in *Proc. Int. Conf. Adv. Technol. Commun. (ATC)*, Hanoi, Vietnam, Oct. 2016, pp. 237–242. [Preprint](#).
- [C35] S. Yang, **K.-H. Ngo**, and M. Kobayashi, “Content delivery with coded caching and massive MIMO in 5G,” in *Proc. Int. Symp. Turbo Codes Iterative Inf. Process. (ISTC)*, Brest, France, Sep. 2016, pp. 370–374. [Preprint](#).
- [C36] **K.-H. Ngo** and Quoc-Tuan Nguyen, “Implementation of network coding scheme in Universal Software Radio Peripheral,” in *Proc. IEICE Int. Conf. Integr. Circuits Design Verification (ICDV)*, Hanoi, Vietnam, Nov. 2014. [Preprint](#).
- [C37] Thai-Mai Dinh Thi, Quoc-Tuan Nguyen, and **K.-H. Ngo**, “Implementation of spectrum sensing scheme in software-defined radio testbed,” in *Proc. IEICE Int. Conf. Integr. Circuits Design Verification (ICDV)*, Hanoi, Vietnam, Nov. 2014.

Conference Posters

- [Po1] F. Fourati, A. U. Rahman, **K.-H. Ngo**, E. J. Oughton, A. Jindal, and M.-S. Alouini, “Optimal network deployment for global connectivity,” in *Eur. Conf. Netw. Commun. and 6G Summit (EuCNC/6G Summit)*, Grenoble, France, Jun. 2022.
- [Po2] **K.-H. Ngo**, “Age of information in prioritized random-access,” in *IEEE SPS - EURASIP Summer School on “Defining 6G: Theory, Applications, and Enabling Technologies”*, Linköping, Sweden, Aug. 2022.
- [Po3] **K.-H. Ngo**, “Massive uncoordinated access for the Internet of Things: A novel information theoretic framework,” in *9th Heidelberg Laureate Forum (HLF)*, Heidelberg, Germany, Sep. 2022.
- [Po4] **K.-H. Ngo**, S. Yang, and M. Guillaud, “Generalized Gaussian model for data-driven learning in communications,” in *Int. Zurich Seminar Inf. Commun. (IZS)*, Zurich, Switzerland, Feb. 2020. [Preprint](#).
- [Po5] T. T. Q. Tran, V.-L. Nguyen, **K.-H. Ngo**, L.-T. Nguyen, Q.-T. Nguyen, E. Bastug, S. Azarian, M. Debbah, and P. Duhamel, “Network coding and information security in Industry 4.0,” in *1st ASEAN IVO Workshop on Cybersecurity and Information Security in Industry 4.0*, Hanoi, Vietnam, Mar. 2019.

Technical Report

- [R1] **K.-H. Ngo**, “Solar panel charge controller,” Dept. of Electrical and Computer Engineering, National University of Singapore, Singapore, Internship report, Aug. 2012.